

TABLE I
Accelerometer Products

Company	Part Number	Output	Dynamic Range	Sensitivity	Frequency Response	Sensitivity Axis	Weight	Size
Analog Devices	ADXL377	Analogue	±200 gm	6.5 mV/gm	1.6kHz	X, Y, Z		3×3×1.45mm
Analog Devices	ADXL375	Digital-SPI & I ² C	±200 gm	20.5 LSB/gm	1.6kHz	X, Y, Z	30 mg	3×5×1mm
STMicroelectronics	LIS32AX	Analogue	±2 gm	336 mV/gm	2.0kHz	X, Y, Z	30 mg	3×3×1.0mm
STMicroelectronics	ALS326DQ	Digital	±2 gm, ±6 gm	±2 g- 1024 LSB/gm ±6 g- 340 LSB/gm	640Hz	X, Y, Z	0.2 gm	7×7×1.8mm
Freescale Semiconductor	MMA1211D	Analogue	±169 gm	13.33 mV/gm	400Hz	Z		
Freescale Semiconductor	MMA16xxKW	Digital	±50 gm - ±312.5 gm	10.24 LSB/gm, 8.192 LSB/gm, 4.096 LSB/gm, 2.731 LSB/gm, 1.638 LSB/gm	400Hz	Z		
Bosch Sensortec	BMA222E	Digital	±2 gm, ±4 gm, ±8 gm, ±16 gm	±2gm-64 LSB/gm ±4gm-32 LSB/gm ±8gm-16 LSB/gm ±16gm-8 LSB/gm	8-1000Hz	X, Y, Z		2×2×0.95mm
Bosch Sensortec	BMA250	Digital	±2 gm, 4 gm, 8 gm, 16 gm	256LSB/gm, 128LSB/gm, 64LSB/gm, 32LSB/gm	8H-1kHz	X, Y, Z		2×2×0.95mm
Bosch Sensortec	BM1055	Digital	±2 gm, 4 gm, 8 gm, 16 gm	1024LSB/gm, 512LSB/gm, 256LSB/gm, 128LSB/gm	8Hz-1kHz	X, Y, Z		
Murata Electronics North America	SCA3060-D01-1	Digital	±2 gm	1000 counts/gm	9Hz	X, Y, Z		7.6×3.3× 8.6mm
Murata Electronics North America	SCA1020-D02-1	Analogue, Digital	±1.7 gm	1.2V/gm	50±30Hz	Y, Z	<1.2 gm	9×5×16mm
Measurement Specialties Inc.	1-1001497-0	Analogue	±150 gm	10mV/gm	2Hz-20kHz	X or Y	8 gm (5 cables)	
Measurement Specialties Inc.	3038-0050	Ratiometric	±50 gm	1mV/gm	1kHz			
Kionix Inc.	KXTF9-2050	Digital	±2 gm, 4 gm, 8 gm	64count/gm, 32count/ gm, 16count/gm (8bit), 1024count/gm, 512count/ gm, 256count/gm (12bit)	25Hz	X, Y, Z		3×3×0.9mm
Kionix Inc.	KXCJ9-1008	Digital	±2 gm, 4 gm, 8 gm	1024 count/gm, 512 count/ gm, 256 count/gm, 64 count/gm, 32 count/gm, 16 count/gm	800Hz	X, Y, Z		3×3×0.9mm
Kionix Inc.	KXR94-2050	Analogue	±2 gm	660mV/gm	800Hz	X, Y, Z		5×5×1.2mm
Memsic Inc.	MXR9500MZ	Analogue, Ratiometric	±1.5 gm	500mV/gm	17Hz	X, Y, Z		7×7×1.8mm
Parallax Inc.	28526	Digital – I ² C	±2 gm, 4 gm, 8 gm	64count/gm, 32count/ gm, 16count/gm (8bit), 1024count/gm, 512count/ gm, 256count/gm (12bit)	250Hz	X, Y, Z		12.7×15.2× 11.7mm

the source of non-uniform frequency response and limited operation bandwidth of the accelerometer.

On the other hand, new gyroscopes have been launched by companies like Epson for car navigation applications. KMX61G is the world's first micro-smp magnetic gyro launched in 2013 by Kionix, with an integrated sensor fusion technology. This reduces the current drawn by up to 90 per cent as compared to the other traditional gyros. New components like these would allow product

designers to incorporate the gyro functionality in products that were restricted in the past from the inclusion of gyro due to their high-power consumption.

Enhancing flexibility

When swapping similar components, engineers always know that not all components are created equal and they will always be mismatched. However, mistakes will always happen and sometimes they tend to be the expensive ones.

Let us consider what happened to Apple's iPhone 5S—by switching the accelerometer from a STMicroelectronics LIS331DLH to a Bosch Sensortec BMA220, the phone literally lost its sense of balance. Now companies like Kionix are coming out with performance optimisation tools that allow a designer to manipulate power and noise levels through a downloadable GUI. This helps them tune the sensor to meet their system's potentially unique requirements through precise design parameter choices.